

Heterogeneous Compliance With The FTC Express Scripts Consent Order: Evidence From The First Post-Settlement Insulin Formulary

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Abstract

The February 2026 Federal Trade Commission consent order against Express Scripts (Docket 9437) is the first behavioral antitrust remedy ever imposed on pharmacy benefit manager (PBM) formulary architecture. Section I prohibits Express Scripts from placing the Low-WAC version of a within-manufacturer authorized-generic insulin pair on a less favorable tier than the same-manufacturer High-WAC version, but does not bind CVS Caremark or OptumRx, both still in administrative litigation. We assembled the first publicly enumerated list of Section-I-qualifying within-manufacturer authorized-generic insulin pairs, ingested the four April 1, 2026 Express Scripts Standard Offering PDFs, and characterized post-Implementation-Date compliance on the four manufacturer pairs the rule reaches. Express Scripts' April 2026 National Preferred Formulary achieves tier parity on the Lilly Humalog and Viartis Semglee pairs, excludes the Novo Nordisk NovoLog franchise entirely (Section I moot), and shows a Section-I-violation-direction configuration on the Sanofi Lantus / unbranded insulin glargine pair — the first publicly observable such configuration on this within-manufacturer pair. Rule-based behavioral antitrust enforcement of PBM formulary architecture is observable, partially effective, and amenable to public-data audit; the Sanofi configuration warrants FTC Section V monitoring.

Key Points

What is known: The three largest pharmacy benefit managers (PBMs) process roughly 79 percent of U.S. prescription drug claims and operate the formularies that determine drug tier placement and utilization management. Substantial literature documents that PBM rebate negotiation distorts formulary placement toward High-WAC products to maximize absolute rebate dollars; patients on coinsurance and high-deductible plans absorb the inflated list prices.

What is new: The February 2026 FTC Express Scripts consent order is the first behavioral antitrust remedy ever imposed on PBM formulary architecture. Its asymmetric application provides a sharp regulatory event affecting one PBM and not its two competitors. The first post-settlement standard offering (April 1, 2026) reverses two previously alleged violations (Lilly Humalog 2019; Viatris Semglee 2021/2024), exits the rule’s domain on a third (Novo Nordisk NovoLog), and shows a first publicly observable Section-I-violation-direction configuration on a fourth pair (Sanofi Lantus / unbranded insulin glargine — a pair for which the FTC’s Part 3 complaint did not encode a pre-period Section-I-direction allegation).

What is next: Section V monitoring of ESI’s compliance analysis on the Sanofi pair; medium-run formulary dynamics through Q3 2026 and Q1 2027 refresh cycles; CVS Caremark and OptumRx settlement or adjudication conversion of the design to staggered adoption.

Introduction

The Express Scripts / FTC consent order (Docket 9437) issued in February 2026 is the most significant antitrust intervention into PBM operations in a decade. Its centerpiece — Section I — prohibits Express Scripts (ESI) from placing the Low-WAC version of a within-manufacturer authorized-generic pair on a less favorable formulary tier than the same-manufacturer High-WAC version, with no stricter utilization management on the Low-WAC version. CVS Caremark and OptumRx remain in active administrative litigation under the same docket and are not legally bound by Section I. This paper exploits the asymmetric application — an asymmetric regulatory event affecting one PBM and not its two competitors — as a first observation point for compliance heterogeneity under a categorical formulary rule.

The rebate-driven formulary distortion the consent order targets is well documented. Net insulin prices fell about 8 percent per year over 2012–2019 while list prices continued to grow; voluntarily negotiated commercial discounts (the rebate channel) absorbed 60.5 percent of gross sales by 2019.^{1,2} PBM-retained shares of insulin spending rose 154.6 percent over 2014–2018.³ The mechanism is contractual: PBM formulary tier-placement discretion gives a credible threat of exclusion that manufacturers offset with rebates tied to list price.^{4,5} Patients on

coinsurance and high-deductible plans absorb the inflated list at the counter.^{6,7}

The FTC’s Part 3 administrative complaint, filed September 2024, contains 274 paragraphs; 64 name a specific insulin SKU and zero name a GLP-1 receptor agonist.⁸ Four paragraphs are load-bearing for this paper: ¶117 (2014 Humalog exclusion); ¶145 (2019 Humalog / Insulin Lispro authorized generic exclusion); ¶154 (2021 Semglee dual-WAC violation); ¶246 (2024 Semglee and Tresiba dual-WAC violation). The complaint alleges that ESI repeatedly placed High-WAC versions on preferred tiers while excluding the same-manufacturer Low-WAC counterparts. The consent order’s Section I prohibits exactly this conduct prospectively.

On April 1, 2026 — the first standard-offering publication after the Section I Implementation Date — ESI released four standard-offering documents: the National Preferred Formulary (NPF), NPF Exclusions list, Flex variant Exclusions list, and High Performance Formulary (HPF). These are the first direct observation of post-settlement compliance.

This paper has three contributions. First, we assemble the first publicly available enumeration of within-manufacturer authorized-generic insulin pairs in the United States, with FDA Purple Book BLA crosswalks and primary-source manufacturer press-release documentation. Second, we estimate a pair-level difference-in-differences specification on the consent order’s effect on within-pair tier placement at ESI relative to CVS Caremark and OptumRx. Third — and this is the substantive contribution — we document a heterogeneous pattern of compliance across the four manufacturer pairs the rule reaches, including one publicly observable configuration that, on a plain reading of the rule, does not comply.

Data and Methods

Study Design

We use a one-treated / two-control difference-in-differences design at the (pair $\times$ PBM $\times$ month) level. ESI is treated at the February 2026 consent-order date; CVS Caremark and OptumRx are controls. The unit of observation is the within-manufacturer authorized-generic insulin pair. The outcomes are the within-pair tier gap (`tier_diff` = `low_wac_tier` - `high_wac_tier`) and a strict Low-WAC preference indicator (`low_wac_preferred` = `1[tier_diff < 0]`). Excluded SKUs are coded at a sentinel tier (99); preferred SKUs are coded at tier 2 on the closed-NPF convention.

Pair Construction

A pair qualifies as Section-I-qualifying when (a) the High-WAC and Low-WAC versions share an FDA Purple Book BLA; (b) the FDA NDC labeler segment

is the same on both sides; (c) the dosage form, route, and strength match; (d) the Low-WAC launch is documented by a primary-source manufacturer press release. Seven pairs qualify: Lantus / Sanofi unbranded insulin glargine (vial and SoloStar formulations; BLA 021081; launched 2019-06-12); Humalog / Lilly Insulin Lispro authorized generic (vial and KwikPen; BLA 020563; launched 2019-05-22); NovoLog / Novo Nordisk Insulin Aspart authorized generic (vial and FlexPen; BLA 020986; launched 2020-01-02); and Semglee / Viatriis unbranded insulin glargine-yfgn (BLA 761201; launched 2021-11-16).

Data Sources

Four classes of evidence fill each pair-PBM-month cell. (1) The FTC Part 3 administrative complaint (§117, §145, §154, §246) provides the ESI pre-period treated baseline (2 Section-I cells with both sides observed). (2) Direct ingestion of the four April 1, 2026 ESI Standard Offering PDFs provides the ESI post-period treated cells (56 cells = 7 pairs × 4 variants × 2 sides). (3) Internet Archive Wayback Machine PBM-formulary PDFs provide the control PBM cells (CVS Caremark 60 percent coverage; OptumRx 86 percent coverage on the high-WAC side). (4) The FDA Purple Book and NDC Directory provide the pair-construction crosswalk. All data are public.

Statistical Specification

We estimate a two-way fixed-effects difference-in-differences at the pair-PBM-month level. Express Scripts is the single treated unit; CVS Caremark and OptumRx are control PBMs. The treatment indicator is the interaction of an Express-Scripts dummy with a post-February-2026 indicator. The headline regression is

$$Y_{p,b,t} = \beta \cdot (D_b \times P_t) + \alpha_p + \gamma_t + \delta_b + \varepsilon_{p,b,t},$$

where $D_b = 1$ for ESI and zero otherwise, $P_t = 1$ for $t \geq$ February 2026 and zero otherwise, and α_p , γ_t , δ_b are pair, month, and PBM fixed effects. Standard errors are clustered at the PBM level (CRV1 with G=3). We estimate on the Section-I subsample (7 pairs × 3 PBMs × 47 months, n=502 with both sides observed) and the full pair panel (15 pairs × 3 PBMs × 47 months, n=1,053).

Robustness Program

We report five implemented structural-rigor checks: (1) a leave-one-out sensitivity across all seven Section-I pairs (eAppendix F.4); (2) exact permutation inference across the three PBM clusters (eAppendix F.3); (3) per-variant subsample difference-in-differences across the NPF, Flex Exclusions, and HPF documents; (4) a stratified hand audit of the OptumRx low-WAC control side (eAppendix F.5); and (5) a control-side last-observation-carried-forward characterization (eAppendix F.6). Three additional specifications — Rambachan-Roth honest sensitivity on the pair-level coefficient, a drop-legal-floor restriction on

the Section-I subsample, and a no-authorized-generic placebo pair — are non-applicable given the degenerate pre-period lead vector (two pre-period treated cells, one treated cluster), the absence of legal-floor-flagged pre-period rows in the Section-I subsample, and the absence of within-manufacturer pairs without an authorized-generic sibling that would otherwise support a placebo. The non-applicability dispositions are detailed in eAppendix F.7.

Results

Structured pre/post contrast statistic (descriptive). On the Section-I subsample ($n=502$), the pair-level structured pre/post contrast on `tier_diff` returns a coefficient of -83.08 ; see Table 1. The operative precision statement is the permutation p of $1/3$ across three PBM clusters (the lowest attainable value); the conventional asymptotic CRV1 SE is 4.08 (asymptotic $p = 0.0024$, 95% CI $[-100.64, -65.52]$) and is reported only for transparency. The coefficient summarizes the treated-side pre/post transition between FTC-pleading-encoded violation cells and the observed April 2026 configuration; the control side is parser-conditioned (eAppendix F.5; OptumRx low-WAC parser audit) and all 56 post-period control cells are LOCF-carried (eAppendix F.6; control-side LOCF characterization). We do not interpret it as a counterfactual estimate. On `low_wac_preferred` (Section-I subsample), the coefficient is -0.414 (SE 0.164 , asymptotic $p = 0.128$). Restricting to the Section-I treated cells, the strict-preference indicator is identically zero in both pre-period and post-period; the negative coefficient is driven by within-PBM $\times$ within-pair fixed-effect demeaning rather than by an observed treated-side movement. The near-zero treated-side result on the strict-inequality outcome is by design — Section I compliance is typically met by tier parity (low = high), not by strict Low-WAC preference, and parity makes the strict-inequality indicator false.

Table 1. Structured pre/post contrast statistic (descriptive).

Subsample	Outcome	$\hat{\beta}$	SE (CRV1)	Asymp. p^a	Permutation p^b	95% CI (CRV1)	n
Section-I only	<code>tier_diff</code>	-83.08	4.08	0.0024	$1/3$	$[-100.64, -65.52]$	502
Section-I only	<code>low_wac_preferred</code>	-0.414	0.164	0.128	$1/3$	$[-1.120, +0.292]$	502
All pairs	<code>tier_diff</code>	-106.20	1.69	0.0003	$1/3$	$[-113.47, -98.92]$	1,053
All pairs	<code>low_wac_preferred</code>	-0.466	0.110	0.713	$1/3$	$[-0.426, +0.519]$	1,053

- (a) Asymptotic CRV1 p , mechanically biased toward over-rejection at $G=3$ with one treated cluster. (b) Exact permutation $p = 0.333$ ($= 1/3$); this

is the lowest attainable value across the three possible PBM relabelings ($G=3$). (c) Per the OptumRx low-WAC parser audit (eAppendix F.5) and the control-side LOCF characterization (eAppendix F.6), the control side is parser-conditioned and LOCF-carried throughout the post-period; the coefficient is best read as a structured pre/post contrast on the treated side rather than as a counterfactual estimate. The per-pair configuration pattern in Table 2 is the substantive empirical contribution.

The magnitude of -83.08 reflects the encoding sentinel (excluded = 99; preferred = 2) and the fact that the FTC Part 3 complaint encoded ESI’s two pre-period cells at `tier_diff` = +97 (full violation). One of those two pre-period cells is from the UNVERIFIED Tresiba pair. The sign is consistent with the direction Section I requires; the magnitude should be read as a direction-of-compliance summary rather than as a calibrated tier-distance effect or a credible causal estimate.

Per-pair compliance pattern (Table 2). The observed per-pair configuration pattern is the substantive contribution of the paper. ESI’s April 2026 NPF achieves Section I tier parity on the Lilly Humalog / Insulin Lispro pair (both sides preferred) and on the Viatris Semglee dual-WAC pair (both sides preferred); the Flex variant is stricter on both pairs (excludes the High-WAC branded version; preserves the Low-WAC counterpart at preferred). The Novo Nordisk NovoLog rapid-acting franchise is excluded entirely on both the NovoLog branded and unbranded Insulin Aspart sides, rendering Section I moot for that manufacturer pair. The Sanofi Lantus / unbranded insulin glargine pair on the standard NPF shows a Section-I-violation-direction configuration: Lantus is preferred (tier 2) and the Sanofi unbranded insulin glargine is excluded. We emphasize that this is the *first publicly observable* Section-I-violation-direction configuration on the Sanofi within-manufacturer pair — the FTC Part 3 complaint did not encode a pre-period Section-I-direction allegation on this pair (§111 alleges Lantus exclusion altogether, which is not the same-manufacturer High-preferred / Low-excluded pattern Section I targets).

Table 2. April 2026 ESI per-pair observed configuration, by formulary variant. “Violation direction” = High-WAC preferred / Low-WAC excluded; “Parity” = both at the same preferred tier (legal-floor compliance); “Strict” = Low-WAC preferred / High-WAC excluded (over-compliance); “Moot” = both excluded (out of Section I’s domain).

Pair	NPF	Flex Excl	NPF Section I status
Lantus / Sanofi unbranded glargine	High preferred / Low excluded	Both excluded	Violation direction
Humalog / Lilly Insulin Lispro	Both preferred	High excluded / Low preferred	Parity (NPF) / Strict (Flex)

Pair	NPF	Flex Excl	NPF Section I status
NovoLog / Novo Insulin Aspart	Both excluded	Both excluded	Moot (manufacturer line dropped)
Semglee / Insulin Glargine-yfgn	Both preferred	High excluded / Low preferred	Parity (NPF) / Strict (Flex)

The Sanofi pair on the April 2026 NPF shows the same Section-I-violation-direction configuration the FTC Part 3 complaint alleged for the Humalog pair in 2019 (§145) and the Semglee pair in 2021/2024 (§154, §246) — but the Sanofi pair itself has no FTC-pleading-encoded pre-period Section-I-violation-direction allegation, so the April 2026 configuration is a first publicly observable violation-direction configuration on this specific pair, not a preserved or reverted one. The configuration is visible on a free PDF available from ESI’s public portal.

Per-pair pre-period evidence base note. Lilly Humalog: FTC §117 (2014), §145 (2019). Viatris Semglee: FTC §154 (2021), §246 (2024). Novo NovoLog: FTC §110 (2023, context only) — the only Section-I-violation-direction allegation on a Novo pair is §246 Tresiba, which is UNVERIFIED and not in the Section-I subsample. Sanofi Lantus: no Section-I-violation-direction allegation; §111 (2023) alleges Lantus exclusion altogether.

Robustness. Placebo on no-AG pairs is uninformative because no Low-WAC counterpart exists. Dropping `source = 'ftc_consent_order_assumed'` rows leaves the headline within the CRV1 CI of Table 1. Per-variant subsample DiDs confirm that the Flex variant pulls more strongly toward compliance than the NPF (= −98.2 on Flex Exclusions vs. = −78.6 on the NPF). Permutation inference across the three PBM clusters yields the lowest attainable p-value of 1/3; the observed coefficient is the most extreme of the three counterfactuals (the CVS and OptumRx counterfactual coefficients are near zero).

Figure references. Figure 1: pair-level mean tier-gap event-study (`figures/fig02.png`). Figure 2: April 2026 per-pair × per-variant compliance grid (`figures/fig03.png`). The pair-PBM-era tier-gap coverage heatmap previously included in the main text is relocated to eAppendix G (eFigure G.1). See `figure-captions.md` for full captions. The main-text exhibit set is Table 1, Table 2, Figure 1, and Figure 2 (four exhibits, at the HAS cap).

Discussion

The consent order moves three of four manufacturer pairs in the direction Section I requires. The Lilly Humalog pair and the Viatris Semglee pair — both

alleged violation sites in the FTC Part 3 complaint — show tier parity on the NPF and strict Low-WAC preference on the Flex variant. The Novo Nordisk NovoLog franchise is dropped entirely from the NPF, which is itself a major formulary action: in 2023 ESI’s NPF preferred Novo’s rapid-acting line per FTC ¶110, and in 2024 ESI’s NPF preserved the High-WAC-preferred / Low-WAC-excluded configuration per ¶246. Exit from the formulary takes the Novo pair out of Section I’s domain.

The Sanofi pair is the outlier. The April 2026 NPF preserves the violation-direction configuration: branded Lantus preferred, Sanofi unbranded insulin glargine excluded. Two interpretive frames bound the finding. One: ESI has read the rule narrowly, treating Sanofi’s Winthrop U.S. generics arm (the labeler of the unbranded version) as a distinct corporate manufacturer for Section I purposes; under this frame, the rebate-gradient interpretation does not apply because the configuration is a rule-design artifact rather than a behavioral response. The other: the configuration reflects deliberate non-compliance, perhaps reflecting a rebate negotiation Sanofi sustained through the consent-order window; under this frame, the rebate-gradient interpretation applies directly. The two interpretive frames map onto distinct readings of the rebate-channel literature.^{1,3,4} The narrow-reading frame is consistent with formulary architecture as a rebate-extraction tool that responds only to enforceable categorical constraints; the deliberate-non-compliance frame is consistent with PBM formulary architecture as a contractually negotiated object that retains rebate-channel flexibility even under behavioral remedy. We cannot adjudicate between these frames with public data and we make no academic claim about which holds. The policy implication is the same under either: the first publicly observable post-settlement standard offering contains a configuration that, on a plain reading of Section I, does not comply. The Novo Nordisk franchise exit, similarly, is consistent with at least three counterfactuals (Section-I-induced exit; pre-existing rebate-contract breakdown; multi-year trajectory toward exclusion of Novo’s rapid-acting line per FTC ¶110); we cannot adjudicate these either, though the observation of franchise exit is robust regardless of cause.

Limitations. Four structural caveats bound the pre/post contrast. (1) The Section-I ESI pre-period treated baseline rests on two FTC Part 3 complaint cells (¶117 / ¶145 on the Humalog pair); Tresiba is `ftc_section_i_pair = UNVERIFIED` and not in the Section-I subsample. A leave-one-out sensitivity (Appendix F.4) shows the Section-I headline coefficient is invariant to dropping Tresiba, and that the headline survives every Section-I per-pair drop (between −82.0 and −89.4). (2) Only one of three PBMs is treated, so conventional cluster-robust inference is mechanically biased toward over-rejection; the permutation p of 1/3 is the operative precision statement. (3) CRV1 inference on three PBM clusters is below the conventional minimum of $G=8$ for credible asymptotic precision; wild-cluster bootstrap and Conley-Taber CRVE are degenerate at $G=3$. (4) The OptumRx low-WAC coverage at 100 percent is a parser artifact: a 10-snapshot stratified hand audit (Appendix F.5) confirms that 21 percent of audited cells trace to concatenation artifacts (e.g., “INSULIN LISPRO 1 UNIT

ISOSORBIDE MONONITRATE160 187 JUNEL”) and 70 percent are LOCF-carried from snapshots whose parsed_rows contain no genuine entry; the 8.6 percent of confirmed-genuine cells all trace to client-specific or state-Medicaid select formularies, not OptumRx’s commercial Premium Formulary. The audit-derived flag has been added to the cleaned panel. The companion control-side LOCF characterization (Appendix F.6) shows that zero of the 56 post-period CVS + OptumRx Section-I cells are directly observed in the underlying SKU panel; every post-period control observation is LOCF-carried, and the headline DiD restricted to directly-observed control rows is mechanically collinear. The implication is that the -83.08 coefficient is best read as a structured pre/post contrast on the treated side with the control side acting as a parser-conditioned reference, not as a counterfactual estimate in the standard sense; the substantive contribution is the per-pair compliance pattern in Table 2, which does not depend on the parser convention. We also acknowledge the short post-treatment window (April–May 2026; the April 2026 configuration is one publication of one PBM and not an equilibrium characterization), the UNVERIFIED Tresiba pair, and the non-adjudicated status of the FTC pleadings. We do not report equity-relevant heterogeneity by patient demographics or plan-design because the source documents (PBM Standard Formularies) carry no beneficiary-level information; downstream out-of-pocket exposure analysis under the consent order requires either a payer-level claims linkage or a plan-sponsor employer-side survey, neither of which is yet feasible at the two-month post-Implementation horizon.

Conclusion

The consent order moves three of four manufacturer pairs in the direction Section I requires; one pair (Sanofi Lantus / unbranded insulin glargine) shows a publicly observable Section-I-violation-direction configuration on the standard NPF. Rule-based behavioral antitrust enforcement of PBM formulary architecture is observable, partially effective, and amenable to public-data audit. The Sanofi pair configuration is the kind of policy signal a rule-based remedy is designed to surface; the consent order’s Section V monitoring framework is in a position to call Express Scripts’ compliance analysis on this specific pair. The medium-run dynamics — Q3 2026 and beyond — and the CVS Caremark / OptumRx settlement trajectory will determine whether the heterogeneity of compliance constitutes a policy success or a regulatory inflection point.

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